

37. Normal Bases in Fields without Higher Ramification

Journal f. d. reine u. angew. Math. 167 (1932), pp. 147–152

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By a normal basis of a Galois number field K/k one means a basis, consisting of conjugate elements, of the maximal order $\mathfrak{O}$ of K with respect to the maximal order $\mathfrak{o}$ of k . A necessary condition for the existence of such a basis is, as is well known,¹ that no higher ramification fields occur in K/k . This remains true if one restricts oneself to one place, that is, if one passes to the $\mathfrak{p}$ -adic extension $k_{\mathfrak{p}}$ of k and to the corresponding $K_{\mathfrak{p}}$ of K . I prove below the validity of the converse:

For all places $\mathfrak{p}$ for which $\mathfrak{p}$ does not divide the degree of K/k , there exists a normal basis. In particular, if K/k has no higher ramification, then a normal basis exists at every ramified place; there the discriminant becomes the square of a group determinant.

To this latter assertion one must add that – contrary to a conjecture of E. Artin – passing to his conductors² still requires further considerations: the decomposition of the group determinant according to irreducible characters gives generalized root numbers; a suitable grouping gives a decomposition in the ground field enlarged by the characters. In the simplest cases I can identify these new factors with Artin's conductors, by means of the ideal-theoretic decomposition of the root numbers.³ I should also note that even in the general case, where no normal basis exists, a splitting into generalized root numbers is always possible by means of a “composition series by Galois modules”, and that this again gives, analogously to the above, a decomposition in the enlarged ground field. Here ideal-theoretic questions begin again.

The proof for the existence of the normal basis under the preceding hypotheses rests on the fact that $\mathfrak{O}/\mathfrak{o}$, regarded as a Galois module – that is, as an $\mathfrak{o}$ -module with the substitutions of the Galois group as operator domain – becomes operator-isomorphic to a (one-sided) ideal of the integral group ring enlarged by $\mathfrak{o}$. At all ordinary ramification places this group ring is a maximal order; ideals in maximal orders of $\mathfrak{p}$ -adic semisimple systems are principal ideals,⁴ and so here they arise from one element by the substitutions of the group, or of the group ring. Translating this back to the Galois module gives exactly the existence of the normal basis. At the higher ramification places the integral group ring passes into a nonmaximal order, and the ideal associated with $\mathfrak{O}/\mathfrak{o}$ passes into a nonprincipal ideal.

All these considerations plainly remain valid for function fields of one variable, provided that the characteristic of the constant field does not divide the degree of K/k .

§1. The $\mathfrak{p}$ -adically extended integral group ring

Let $\mathfrak{o}$ and $\mathfrak{o}_{\mathfrak{p}}$, respectively, denote the maximal orders of an algebraic number field k and of its $\mathfrak{p}$ -adic extension $k_{\mathfrak{p}}$, where $\mathfrak{p}$ is a prime ideal of k . Further, let $(\mathfrak{G})$ denote the rational group ring and $[\mathfrak{G}]$ the ordinary integral group ring of a group $\mathfrak{G}$ of order n . By $(\mathfrak{G})_k$, respectively $(\mathfrak{G})_{k_{\mathfrak{p}}}$, we mean the group ring with coefficients in k , respectively in $k_{\mathfrak{p}}$; correspondingly $[\mathfrak{G}]_{\mathfrak{o}}$, respectively $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$, denotes the extension of the integral group ring to one with coefficients in $\mathfrak{o}$, respectively in $\mathfrak{o}_{\mathfrak{p}}$.

Thus $(\mathfrak{G})_k$ and $(\mathfrak{G})_{k_{\mathfrak{p}}}$ are systems without radical (semisimple systems), and $[\mathfrak{G}]_{\mathfrak{o}}$, respectively $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$, are orders in $(\mathfrak{G})_k$, respectively in $(\mathfrak{G})_{k_{\mathfrak{p}}}$.

Theorem 1. *If the prime ideal $\mathfrak{p}$ does not divide the order n of $\mathfrak{G}$, then $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ is a maximal order of $(\mathfrak{G})_{k_{\mathfrak{p}}}$; if $\mathfrak{p}$ divides n , then $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ is not maximal.*

¹Cf. A. Speiser, Gruppensdeterminante und Körperdiskriminante, § 6. Math. Ann. 77 (1916), pp. 546–562.

²E. Artin, Über die gruppentheoretische Struktur der Diskriminante. J. f. Math. 164 (1931), pp. 1–11.

³[6 Oct. 1931.] That ideal-theoretic considerations will still be necessary in general is shown by the following example, communicated to me by M. Deuring and variable at will, of a nonmaximal order whose discriminant can be represented as the square of the group determinant, while different ideal factors correspond to conjugate characters. Let k be the field of fifth roots of unity, $K = k(\sqrt[5]{2})$. Consider the order $[1, 2\sqrt[5]{2}, \sqrt[5]{2^2}, \sqrt[5]{2^3}, \sqrt[5]{2^4}]$, at the place (2), where by the considerations of this note it has a normal basis. The decomposition according to the characters gives precisely the above basis elements as factors for the corresponding group determinant. Thus the “conductors”, apart from 1, become $2\sqrt[5]{2} \cdot \sqrt[5]{2^4}$, $\sqrt[5]{2^2} \cdot \sqrt[5]{2^3}$, $\sqrt[5]{2^3} \cdot \sqrt[5]{2^2}$, $\sqrt[5]{2^4} \cdot 2\sqrt[5]{2}$, that is, 4, 2, 2, 4; hence they are different for conjugate characters.

⁴Cf. H. Hasse, Über $\mathfrak{p}$ -adische Schiefkörper und ihre Bedeutung für die Arithmetik hyperkomplexer Zahlssysteme, Theorems 46 and 47. Math. Ann. 104 (1931), pp. 495–534. For commutative systems the principal-ideal property already holds after passing to the quotient ring with respect to $\mathfrak{p}$ in k ; one may therefore restrict to this weaker extension for the definition of the place when Abelian groups and fields are involved.

The discriminant of the integral group ring, and hence also that of $[\mathfrak{G}]_{\mathfrak{o}}$ and of $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$, is a power of the order n .⁵ Thus if $\mathfrak{p}$ does not divide n , the discriminant of $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ is a unit. This means that, while keeping $\mathfrak{o}_{\mathfrak{p}}$ fixed, $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ cannot be enlarged to a more comprehensive order; for such an enlargement would correspond to splitting off a quadratic nonunit factor from the discriminant. But $\mathfrak{o}_{\mathfrak{p}}$ was already assumed to be the maximal order, hence maximal in $k_{\mathfrak{p}}$. This proves the first part of Theorem 1, the only part used below.

If, on the other hand, $\mathfrak{p}$ divides n , then the direct sum corresponding to the identity representation

$$\mathfrak{C} = E^{(1)}[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}} + (1 - E^{(1)})[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}, \quad E^{(1)} = \frac{1}{n} \sum_{S \in \mathfrak{G}} S,$$

is a proper enlargement of $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$. Indeed, because $E^{(1)} = \frac{1}{n} \sum S$, $\mathfrak{C}$ is a proper enlargement; $\mathfrak{C}$ is an order with the dependent generators $E^{(1)}$, $(1 - E^{(1)})S$ with $S \in \mathfrak{G}$ ($E^{(1)}S = E^{(1)}$).

Theorem 2. *If $\mathfrak{p}$ does not divide n , then every integral or fractional ideal with respect to $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ is a principal ideal.*

For by Theorem 1, $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ is a maximal order in a $\mathfrak{p}$ -adic semisimple system; its ideals are therefore principal ideals.⁶

§2. Galois modules, operator-isomorphism, normal basis

Let K/k be Galois and let $\mathfrak{G}$ be its group; z^S denotes the element of K obtained from z by the substitution S of $\mathfrak{G}$. The substitutions of $\mathfrak{G}$ produce operator automorphisms on K/k , considered as a k -module, that is, as an additive Abelian group with k as operator domain. Hence K/k may be made a module with respect to $(\mathfrak{G})_k$ by setting

$$z \left(\sum_i S_i c_i \right) = \sum_i z^{S_i} c_i, \quad z \in K, \quad S_i \in \mathfrak{G}, \quad c_i \in k.$$

Definition. *Every $(\mathfrak{G})_k$ -module from K/k is called a rational Galois module. Likewise the $[\mathfrak{G}]_{\mathfrak{o}}$ -modules from K/k are called integral Galois modules; in particular $\mathfrak{D}/\mathfrak{o}$ is an integral Galois module.*

Theorem 3. *As a rational Galois module, K/k is operator-isomorphic to $(\mathfrak{G})_k$.*

This follows from the fact that K/k always has a field normal basis, that is, a k -basis consisting of conjugate elements z^S .⁷ The correspondence

$$S \longmapsto z^S, \quad \sum_i S_i c_i \longmapsto \sum_i z^{S_i} c_i \quad (c_i \in k), \quad \text{hence } ST \longmapsto z^{ST},$$

gives an operator homomorphism, which becomes an isomorphism because the ranks over k are equal.

Theorem 4. *As an integral Galois module, $\mathfrak{D}/\mathfrak{o}$ is operator-isomorphic to a $[\mathfrak{G}]_{\mathfrak{o}}$ -module from $(\mathfrak{G})_k$, of highest rank n . Likewise $\mathfrak{D}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ is operator-isomorphic to a $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ -module of rank n from $(\mathfrak{G})_{k_{\mathfrak{p}}}$.*

The $(\mathfrak{G})_k$ -isomorphism $K/k \rightarrow (\mathfrak{G})_k$ given in Theorem 3 assigns to the $[\mathfrak{G}]_{\mathfrak{o}}$ -module $\mathfrak{D}/\mathfrak{o}$ a $[\mathfrak{G}]_{\mathfrak{o}}$ -module of rank n in $(\mathfrak{G})_k$. Further, the $(\mathfrak{G})_k$ -isomorphism extends to a $(\mathfrak{G})_{k_{\mathfrak{p}}}$ -isomorphism from $K_{\mathfrak{p}}/k_{\mathfrak{p}}$ to $(\mathfrak{G})_{k_{\mathfrak{p}}}$, by allowing the coefficients c_i in the correspondence of Theorem 3 to range over all elements of $k_{\mathfrak{p}}$. This isomorphism then induces the $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ -isomorphism of $\mathfrak{D}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$. Here $K_{\mathfrak{p}}/k_{\mathfrak{p}}$ is to be regarded as a hypercomplex system over $k_{\mathfrak{p}}$. It arises from K/k by extension of coefficients; in general it becomes a system with zero divisors, namely a sum of (isomorphic) fields, and hence a semisimple system.

⁵Cf. for example E. Noether, Hyperkomplexe Größen und Darstellungstheorie, § 26. Math. Ztschr. 30 (1929), pp. 641–692.

⁶Hasse, loc. cit. The principal-ideal property is proved there only for simple systems, but by standard arguments it follows also for semisimple ones. The components of a maximal order are again maximal; hence the components of the ideal under consideration are principal ideals, say with bases a_i . Then $a = \sum a_i$ is a basis of the original ideal. For Abelian groups compare also note 3.

⁷If $a_1, \dots, a_n$ is a basis of K/k and the u_i are indeterminates, then the $\sum u_i a_i^S$, as S runs through the elements of $\mathfrak{G}$, are linearly independent; hence the u_i may be specialized to elements of k in such a way that the linear independence is preserved.

Theorem 5. *At every place $\mathfrak{p}$ which does not divide the degree n of K/k , $\mathfrak{O}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ has a normal basis.⁸*

For by Theorem 1, $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ is a maximal order there; the $[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$ -modules of rank n from $(\mathfrak{G})_{k_{\mathfrak{p}}}$ therefore become integral or fractional ideals, and principal ideals by Theorem 2. Thus if the ideal associated with $\mathfrak{O}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ is equal to $W[\mathfrak{G}]_{\mathfrak{o}_{\mathfrak{p}}}$, then it has the $\mathfrak{o}_{\mathfrak{p}}$ -basis $W S_i$. The operator-isomorphism says that w^{S_i} is an $\mathfrak{o}_{\mathfrak{p}}$ -basis of $\mathfrak{O}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$, where w denotes the element of $\mathfrak{O}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ corresponding to W . The n conjugate elements w^{S_i} therefore form a normal basis of $\mathfrak{O}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$.

If $\mathfrak{p}$ divides n , then the module associated with $\mathfrak{O}_{\mathfrak{p}}/\mathfrak{o}_{\mathfrak{p}}$ cannot be principal, since in that case no normal basis can exist.

Additional remark to Theorem 3. From the operator-isomorphism of $(\mathfrak{G})_k$ with K/k proved in Theorem 3⁹ follow Speiser's results on Klein's problem of forms,¹⁰ which can be formulated as follows:

Every representation of the Galois group $\mathfrak{G}$ of K/k irreducible over k is produced by an irreducible (rational) Galois module from K/k ; every absolutely irreducible representation is produced by a Galois module from K_Z/Z .

Here Z denotes a splitting field, containing k , of the corresponding component of the group ring; K_Z/Z is to be regarded as a hypercomplex system over Z , arising by extension of coefficients from K/k . In particular K_Z remains a field if Z can be chosen relatively prime to K over k , so that K_Z/Z becomes an accessory extension, the case treated by Speiser.

The assertion itself follows, by means of operator-isomorphism, from the corresponding assertion for $(\mathfrak{G})_k$ or $(\mathfrak{G})_Z$, where the irreducible ideals produce the representation.

If $v_1, \dots, v_t$ is a basis of such a Galois module, and $S \mapsto \bar{S}$ is the corresponding representation, then

$$\begin{pmatrix} \vdots \\ v_i^S \\ \vdots \end{pmatrix} = \bar{S} \begin{pmatrix} \vdots \\ v_i \\ \vdots \end{pmatrix}.$$

This is Klein and Speiser's formulation.

§3. The discriminant as group determinant in fields without higher ramification

For decomposing the discriminant it is known to be enough to consider the decomposition at the individual ramification places; for fields without higher ramification, only places with a normal basis are involved.

Theorem 6. *For Galois fields K/k without higher ramification, the discriminant at every ramification place is equal to the square of the group determinant of the Galois group, with the indeterminates replaced by the normal basis. Corresponding to the irreducible representations, the group determinant decomposes into generalized root numbers; the discriminant decomposes into ideals of the ground field enlarged by the characters, with the same ideals corresponding to conjugate-complex characters.*

For with w^S as normal basis the discriminant Δ is the square of

$$D = |w^{ST^{-1}}|; \quad S, T \in \mathfrak{G}.$$

Thus D is the group determinant with w^S in place of the indeterminates x_S .

The decomposition into root numbers follows from Speiser's arguments, loc. cit. Let

$$D = D_1^{f_1} \cdots D_t^{f_t}, \quad M = f_1 M_1 + \cdots + f_t M_t$$

be the decomposition of the group determinant, respectively the group matrix, corresponding to the absolutely irreducible representations. If $S \mapsto \bar{S}$ is the representation corresponding to M_λ , where the $\bar{S}$

⁸In the case of Abelian fields, the place may also be defined by the quotient ring, as in notes 3 and 5.

⁹The proof plainly assumes only that K is Galois of the first kind over k , and that k has infinitely many elements. This last restriction is unnecessary: M. Deuring has found a proof of Theorem 3 valid also for fields with finitely many elements, from which conversely the existence of the field normal basis follows. At the same time this gives a proof for the existence of the primitive element, running uniformly for fields with finitely and infinitely many elements.

¹⁰Loc. cit. The concept of the Galois module is abstracted from there. If the characteristic of k divides n , then one is dealing with composition series by Galois modules and their irreducible factors.

lie in the enlarged ground field, then

$$M_\lambda = \sum_S w^S \bar{S}, \quad M_\lambda^{T^{-1}} = \sum_S w^{ST^{-1}} \bar{S} = \sum_R w^R \bar{R} \bar{T} = M_\lambda \bar{T},$$

and passing to determinants gives

$$D_\lambda^{T^{-1}} = D_\lambda |\bar{T}| = D_\lambda \varepsilon_T.$$

Thus the D_λ are recognized as generalized root numbers; ε_T , being the determinant of $\bar{T}$, is a root of unity (an irreducible representation of the factor group of $\mathfrak{G}$ by its commutator group). For cyclic groups the D_λ are simply the Lagrange root numbers $\sum w^S \chi_\lambda(S)$.

In general the D_λ lie in the hypercomplex system obtained from $K_{\mathfrak{p}}/k_{\mathfrak{p}}$ by enlarging the coefficient field $k_{\mathfrak{p}}$ with the corresponding character; indeed they are integral quantities of this system. For the determinant formed with indeterminates x_S has coefficients which are algebraic integers from the character field, whereas the w^S are integral elements of $K_{\mathfrak{p}}$.

To pass from the decomposition of the group determinant to that of the discriminant, each representation is combined with its adjoint. If $\lambda, \bar{\lambda}$ denote adjoint representations, then simultaneously

$$D_\lambda^{T^{-1}} = D_\lambda \varepsilon_T, \quad D_{\bar{\lambda}}^{T^{-1}} = D_{\bar{\lambda}} \varepsilon_T^{-1};$$

at the same time the determinants belonging to λ and $\bar{\lambda}$, when formed with indeterminates x_S , are conjugate-complex, while in general conjugate determinants correspond to conjugate characters for indeterminates x_S .

Putting $\Delta_\lambda = D_\lambda D_{\bar{\lambda}}$, it is therefore necessary to adjoin only the real subfield of the λ -th character, and

$$\Delta_\lambda^T = \Delta_\lambda \quad \text{for all } T \text{ in } \mathfrak{G};$$

hence¹¹: Δ_λ lies in the ground field enlarged by the real subfield of the λ -th character.

The discriminant decomposition

$$\Delta = \Delta_1^{f_1} \dots \Delta_t^{f_t}$$

is therefore a decomposition in the ground field enlarged by the real subfields of the characters, with the same factors corresponding to conjugate-complex characters. For the other conjugate characters nothing further can be concluded without more [cf. 2a)].

This proves Theorem 6 in all its parts. If it can be shown that the ideals generated by the Δ_λ in fact already lie in the ground field k , and are equal for conjugate characters, then they agree with Artin's conductors, provided one assigns to the composite characters the corresponding power products of the Δ_λ . For both the functional equation and the fact following directly from the normal basis (cf. Speiser, loc. cit.) hold: the discriminants of the corresponding subfields correspond to the identity representations of the subgroups. Agreement at every place then implies agreement of the full ideals. If one restricts to rationally irreducible characters, the stated facts yield the agreement. For the absolutely irreducible characters the agreement is to be confirmed directly in the simplest case:

Theorem 7. *Let k be the field of rational numbers and let K/k be cyclic of prime degree l prime to the discriminant, hence without higher ramification. Then the ideals generated by the Δ_λ for the nonidentity representations are all equal to one another and agree with the conductor of K/k .*

This follows from the known decomposition of the root numbers.¹² At a ramification place $\mathfrak{p}$ of K/k , for k_ε , the field of the l -th roots of unity, and for K_ε (K_ε remains a field, since k_ε is relatively prime to K

¹¹Since this is a hypercomplex coefficient extension of a field, the usual inference of Galois theory must be modified. Let P be the extension field of $k_{\mathfrak{p}}$ under consideration and

$$(K_{\mathfrak{p}})_P = (K_{\mathfrak{p}})_P e^{S_1} + \dots + (K_{\mathfrak{p}})_P e^{S_r}$$

be the direct decomposition into fields. Then $(K_{\mathfrak{p}})_P e^{S_i} / P e^{S_i}$ is Galois (the group is a subgroup of the decomposition group of a prime factor of $\mathfrak{p}$), and the e^{S_i} are conjugate with respect to the cosets. From $\Delta_\lambda^T = \Delta_\lambda$ it follows first that the i -th component of Δ_λ lies in $P e^{S_i}$, say is $\gamma_i e^{S_i}$ with $\gamma_i \in P$. From the conjugacy of the e^{S_i} it follows that all γ_i are equal, so that actually $\Delta_\lambda = \gamma \sum e^{S_i} = \gamma$ lies in P .

¹²Hilbert, Zahlbericht, § 108. Since by Theorem 5 the existence of the normal basis is established – the place may simply be defined by the quotient ring according to note 7 –, the fact used in the Zahlbericht that absolutely cyclic fields are cyclotomic fields need no longer be invoked.

and the $\mathfrak{p}$ -adic extension is superfluous here by note 7), one has

$$(p) = \mathfrak{p}_1 \cdots \mathfrak{p}_{l-1}, \quad \mathfrak{p}_i = \mathfrak{P}_i^l,$$

with $\mathfrak{p}$ in k_ε and $\mathfrak{P}$ in K_ε . Further, for the root numbers,

$$(\Omega_\lambda) = \mathfrak{P}_1^{r_1} \cdots \mathfrak{P}_{l-1}^{r_{l-1}}, \quad (\Omega_{\bar{\lambda}}) = \mathfrak{P}_1^{l-r_1} \cdots \mathfrak{P}_{l-1}^{l-r_{l-1}},$$

where $r_1, \dots, r_{l-1}$ is a permutation of the numbers $1, \dots, l-1$. But the root numbers Ω_λ are here identical with the factors D_λ of the group determinant; therefore

$$(\Delta_\lambda) = (D_\lambda D_{\bar{\lambda}}) = (\Omega_\lambda \Omega_{\bar{\lambda}}) = \mathfrak{P}_1^l \cdots \mathfrak{P}_{l-1}^l = \mathfrak{p}_1 \cdots \mathfrak{p}_{l-1} = (p),$$

hence agreement with the conductor of K/k .

Received 24 August 1931.